

EXPERIMENT 15 – Product Sales Analysis

Question

Analyze monthly product sales using grouped bar chart, stacked area chart, and table.

Dataset

Product	January	February	March
Product A	2000	2200	2400
Product B	1500	1800	1600
Product C	1200	1400	1100

The supplied dataset and visualization requirements are given under Product Sales Analysis.

R Code

```
# Experiment 15 - Product Sales

products <- c("Product A", "Product B", "Product C")

sales <- matrix(c(
  2000, 2200, 2400,
  1500, 1800, 1600,
  1200, 1400, 1100
), nrow = 3, byrow = TRUE)

colnames(sales) <- c("January", "February", "March")
rownames(sales) <- products

# Question 1 - Grouped Bar Chart

barplot(sales,
  beside = TRUE,
  main = "First Quarter Product Sales",
  xlab = "Month",
```

```

    ylab = "Sales",
    col = c("skyblue", "lightgreen", "orange"))
legend("topleft",
    legend = products,
    fill = c("skyblue", "lightgreen", "orange"))
# Question 2 - Stacked Area Chart
months <- 1:3
plot(months, sales[1, ],
    type = "n",
    ylim = c(0, max(colSums(sales))),
    xaxt = "n",
    main = "Overall Sales Trend",
    xlab = "Month",
    ylab = "Sales")
axis(1, at = months,
    labels = colnames(sales))
polygon(c(months, rev(months)),
    c(sales[1, ], rep(0, 3)),
    col = "skyblue")
polygon(c(months, rev(months)),
    c(sales[1, ] + sales[2, ], rev(sales[1, ])),
    col = "lightgreen")
polygon(c(months, rev(months)),
    c(colSums(sales), rev(sales[1, ] + sales[2, ])),
    col = "orange")
# Question 3 - Table

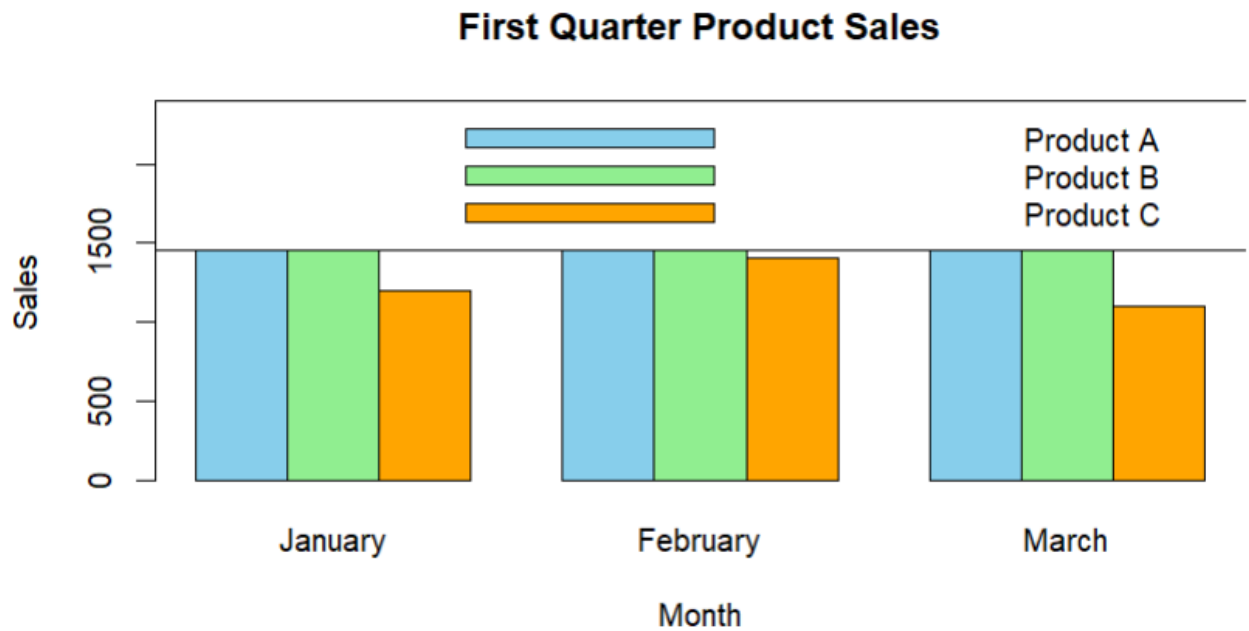
```

```
print(sales)
```

Question 1

Create a grouped bar chart to visualize sales of each product for the first quarter.

Output 1

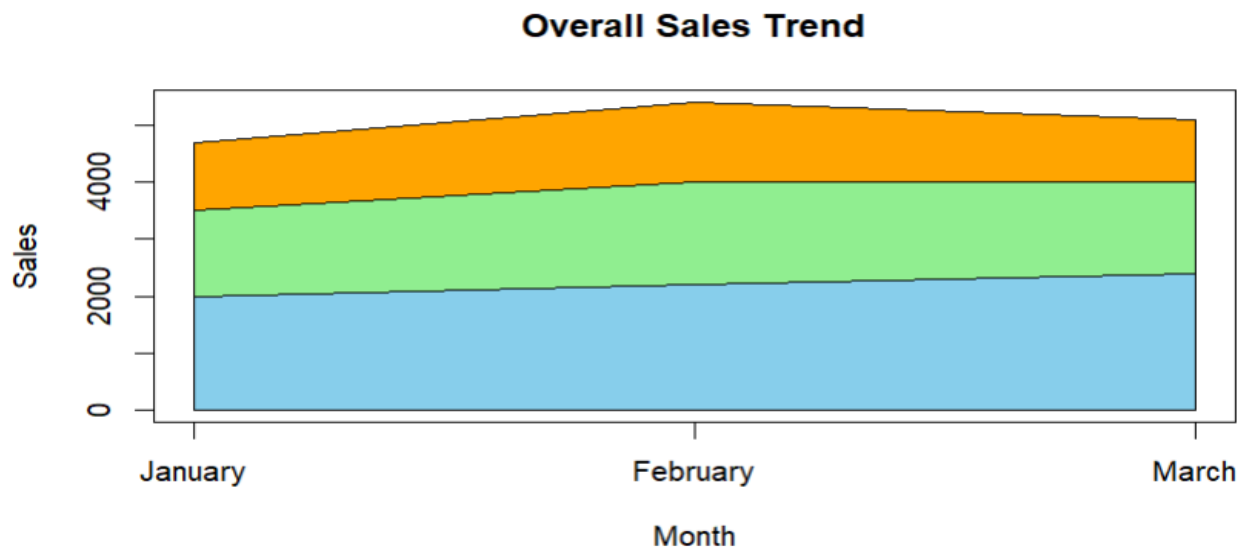


Product A has the highest sales in all three months.

Question 2

Generate a stacked area chart showing overall sales trends.

Output 2



Total monthly sales are:

January = 4700

February = 5400

March = 5100

Sales increase from January to February and slightly decrease in March.

Question 3

Display monthly sales data.

Output 3

```
> # Question 3 - Table
> print(sales)
```

	January	February	March
Product A	2000	2200	2400
Product B	1500	1800	1600
Product C	1200	1400	1100